

DECISION MEMORANDUM

ACTION

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17 April 2023

MEMORANDUM FOR SECRETARY OF STATE ANTONY BLINKEN, SECRETARY OF
DEFENSE LLOYD JAMES AUSTIN III

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SUBJECT: Addressing Fully Autonomous Nuclear Weapons Launch

EXECUTIVE SUMMARY

The rapidly advancing technological development of artificial intelligence and the evolving geopolitical landscape are posing new challenges to the nuclear nonproliferation regime. Specifically, new breakthroughs in artificial intelligence could soon equip nuclear states with the capability for fully autonomous nuclear weapons launch, which could inevitably increase the chance of inadvertent deployment due to miscalculation, technical failure, and cyber-attacks. Meanwhile, both Russia and China are seeking to upgrade their respective nuclear arsenals and incorporate autonomous capabilities into their nuclear configurations, albeit not explicitly on weapons launch. According to the assessment of this report, the current nonproliferation regime is not adequate to address these nascent yet urgent challenges. Therefore, this memo articulates five policy options for the US government to consider: 1) maintaining the status quo and focusing on its own autonomous capability, 2) making unilateral statements or policies to indicate the refusal for autonomous launch, 3) exploring bilateral agreements with Russia and/or China for arms control, 4) pursuing multilateral frameworks to curb the development and deployment of autonomous launch, and 5) a legal-binding treaty option with formal verification requirements. Considering risks, feasibility, international norms, and other factors, this memo recommends prioritizing the fourth option, which is to engage with nuclear states, especially Russia and China, to seek dialogue on the issue and public non-binding commitments against fully autonomous nuclear weapons launch.

BACKGROUND

The nuclear nonproliferation regime has done a remarkable job at limiting the spread of nuclear weapons and related systems globally. These are evident in the absence of use of nuclear weapons in military conflict since 1945, the limited number of nuclear weapons states, and spread of peaceful nuclear technologies and applications. However, rapid technological advancement calls for the adaptation of the current nonproliferation regime. Specifically, the current regime needs to consider emerging technologies and the opportunities and threats they may pose for nuclear nonproliferation. An important field with rapid expansion is autonomy which finds its core architecture in artificial intelligence. Currently, we do not know of any state that has incorporated fully autonomous functionality into their nuclear command and control systems. However, due to recent development in nuclear weapons modernization, and rapidly advancing artificial intelligence capabilities, it has become important to address the impact on the current nonproliferation regime by providing recommendations to address actions needed to be taken on the use of fully autonomous technology for launching nuclear weapons.

A fully autonomous weapon consists of a system that can select and engage targets without human intervention. According to the U.S. Department of Defense (DOD) policy directive on autonomous weapons, the DOD describes the responsibilities and guidelines associated with the development and deployment of semi-autonomous to fully autonomous weapons. Additionally, the DOD announced in February 2020 the establishment of five responsible artificial intelligence (AI) principles that will serve as a platform to autonomous weapons systems. These include responsibility, equitability, traceability, reliability, and governance. Nuclear armed states like Russia and China had also voiced out their interest in nuclear modernization and to incorporate autonomous capabilities in their nuclear weapon systems. President Putin for example has emphasized that “digital technologies and AI, robotization, and unmanned systems should be on the qualitative development agenda of our Armed Forces.” Additionally, China has set goals to further the advancement of AI innovation, where its military weapon manufacturer projected that by 2025, lethal autonomous weapons would be commonplace.

Despite these advancements and the rapid growth of interest in autonomous nuclear weapon systems, there are currently no international frameworks to guide the nonproliferation regime on artificial intelligence (AI)-enabled autonomous systems. Since 2014, the U.S. has engaged internationally on Lethal Autonomous Weapon Systems (LAWS) under the patronage of the United Nations Convention on Certain Conventional Weapons (U.N. CCW). The informal discussions were then transitioned into a formal meeting known as the Group of Governmental Experts (GGE) in 2017 tasked with examining the technological, ethical, military, and legal pathways of LAWS. Between 2018 and 2019, GGE considered proposals by states parties with the aim to issue political declarations about LAWS and regulate development and if applicable, their usage. Concerns such as accountability for use, compliance, and operational risks later boosted the call for the ban of LAWS by about 30 countries and 165 nongovernmental organizations. However, some nuclear states including the United States do not currently support the ban on LAWS. In contrast to the ban of LAWS, the United States has raised ethical concerns

about the humanitarian benefits of the system. This, including declarations by other states has contributed to the stillness of any international efforts for the regulations of autonomous nuclear weapon systems and with fully autonomous capabilities.

POLICY OPTIONS

Given the geopolitical and cyber concerns surrounding the possible emergence of fully autonomous nuclear weapons launch systems, the United States must carefully approach the issue. There are five policy options to confront the risks:

1. Status quo. Under the status quo, the United States will take no action to limit the development of autonomy for our nuclear weapons program. Leaving open the option to develop fully autonomous nuclear weapons launch will allow us to attempt to gain parity with China, seen as the current leader in this area. Permitting free growth in this area could support the emergence of US leadership in autonomy technologies, which has security benefits. This option could lead to an arms race with China, Russia, or others, which could be both costly and dangerous. The quick pace of technological development could pose unforeseen cyber risks or jump ahead of the regulation curve.
2. Unilateral action. The United States could take unilateral action to limit our development of autonomous nuclear launch. We could frame this as an international security issue on which we are taking an individual stand and invite other countries that we see as technologically capable in this area to make similar statements. Unilateral limits are consistent with the Department of Defense's current posture in which only humans make decisions about deploying lethal force. Further, unilateral limits could reduce cyber risks related to integration of overly complex systems into our nuclear weapons launch protocols. Limited autonomy research for nuclear launch could save research funding to be reinvested in more pressing areas such as climate change. However, unilateral action could allow the United State to be technologically overcome by autonomy research in adversary countries.
3. Bilateral agreement. Bilateral agreements could potentially be negotiated with Russia or China, in a similar style to previous bilateral nuclear arms accords. However, given China's historical refusal to reduce nuclear arms until the United States and Russia decrease arsenals to the scale of China's, only Russia is considered in this memo. Despite our history of arms control agreement with Russia, the war in Ukraine and suspension of the New START Treaty make engagement with Russia significantly more difficult than usual. In addition, a bilateral agreement typically could include some type of verification measure, which would pose new technical challenges if computer code is to be verified.
4. Multilateral agreement. A multilateral agreement against the development of fully autonomous nuclear launch could represent an international dialogue on the issue. A multilateral agreement would not likely be legally binding, and could potentially have light or no verification measures, for example based on the annual self-reporting that states submit under their United Nations Security Council Resolution 1540 obligations, or groups of experts for conventional arms control measured under the United Nations. Though a

multilateral agreement could proceed without Russia or China, it is not clear if it would have impact without these key players.

5. Binding treaty. A binding treaty could make illegal the use of full autonomy for nuclear weapons launch. The treaty would be accompanied by formal verification measures carried out by a group of members or a designated third party. A binding treaty would have international legal enforcement, for which a state not complying with the treaty could be sanctioned by the United Nations. The verification measures could build confidence among treaty partners that autonomous launch is not needed. While it is possible to have treaties like this enter into force, unless the United States, Russia, and China all sign and ratify (which in the present is incredibly unlikely due to geopolitical tensions) the treaty will be missing the major players and have limited impact. Furthermore, opening the US command and control system to international inspections would require much more cyber risk than bilateral inspection regimes with a single nuclear-armed partner.

RECOMMENDED COURSE OF ACTION

We recommend pursuing a multilateral agreement between the United States, Russia, and China. Due to the risks of a potential arms race and cyber-attacks infiltrating autonomous nuclear weapon systems, it is necessary to take action to limit this technology in nuclear weapon systems. A multilateral agreement between the United States, China and Russia is the most feasible option within the current geopolitical climate. With current tensions between the United States and Russia or China, a multilateral agreement limiting autonomous nuclear weapons launch will still be challenging to achieve, but we think it is worth the effort given the potential risks.

Unlike binding treaties, which can include strict requirements and invasive verification measures, non-binding multilateral agreements may be more flexible and allow countries to express intent to limit autonomy in a more politically feasible way. A multilateral approach – if the parties follow through with individual commitments or statements to limit their autonomous nuclear launch capabilities, could reduce the cyber security risk of autonomous launch, and limits potential security risks from third party verification activities. A multilateral approach could also open opportunities for countries to work together to develop cybersecurity measures and reduce the risk of cyber-attacks, especially when discussing newer technologies.

A multilateral agreement to limit autonomous nuclear weapon launch will start meaningful conversations, which could lead to better cooperation and understanding between countries. As the technology for autonomous systems continues to advance, it is important for countries to work together to establish rules and regulations around their use. This agreement can provide a forum for these discussions and can help to ensure that the development of autonomous systems is safe and responsible. While binding treaties may be the longer-term goal, multilateral agreements can provide a more practical and productive way of addressing global issues in the short term. By working together and starting meaningful conversations, countries can build trust and cooperation, which can lead to more effective and binding agreements in the future.